

Subscription App Analysis by Marharyta Hryshyna

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Context and Purpose of the Analysis

The product under analysis is a mobile app for learning English, operating on a subscription model (trial → paid → renewal → upgrade). The analysis covers 5,000 users and 23,020 transactions (February 2024 – August 2025), along with marketing spend data by channel.

Two key problems:

- The stage of the funnel (Signup→Trial→Payment→Renewal→Upgrade) responsible for the largest user drop-off had not been identified.
- Channel and country performance in terms of return on acquisition spend (LTV/CAC) had not been assessed.

Objective: to build a complete picture of the funnel and unit economics, identify drop-off points, assess channel and country profitability, and provide recommendations on budget prioritisation and improving trial-to-paid conversion.

Key Findings

- The two largest funnel drop-offs occur at Signup→Trial (23% of users never start a trial) and Renewal→Upgrade (only 21% of renewing users move to a higher-tier plan).
- 32% of paying users (1,150 of 3,602) go straight to first payment without a trial, a distinct conversion path worth targeting separately in marketing.
- Retention stabilises at a healthy 69% from month 4 onward; users who survive the initial drop-off tend to stay long-term.
- The apparent LTV/CAC "paradox" across countries (Brazil ratio 7.7 vs Germany 3.2) largely disappears once unattributed Global marketing spend is allocated proportionally — the ratio converges to 1.8–2.0 for all six countries, indicating the spread was mostly a cost-attribution artefact rather than a real difference in market efficiency.
- Influencer marketing has the lowest CAC among paid channels (\$8.39) and the best LTV/CAC ratio (6.68); Google Ads and Apple Search Ads drive the most volume but at a lower return per dollar spent.

Methodology

The dataset consists of four tables: users (5,000 rows), transactions (23,020 rows), subscription_plans (5 rows), and marketing_spend (aggregated by country, channel and month). Data auditing confirmed table volumes, referential integrity between users and transactions, and business rules (e.g. refunds equal 2% of first payment).

Key metric definitions:

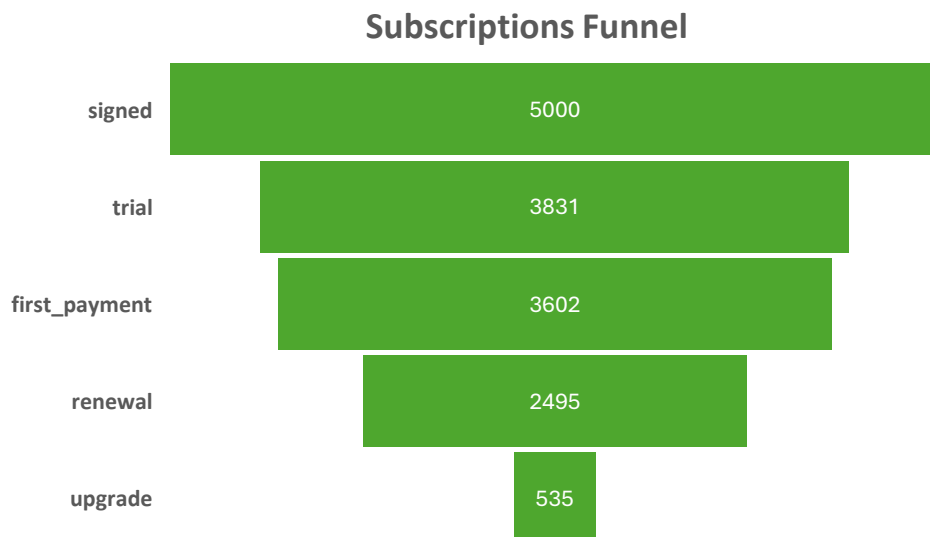
- **LTV** — total revenue per user, blended across all acquisition channels (including free/organic ones).
- **CAC** — total marketing spend divided by new users acquired, calculated per channel and per country; Global (SMM+Content) spend, not attributable to a single country, was allocated proportionally to each country's share of new users where noted.

- **Retention** — monthly cohort analysis, grouping users by signup month and tracking activity through M+11; annual/lifetime plan holders are tracked via a snapshot date (31 Aug 2025) rather than individual transaction dates.

Analysis was carried out in SQL (MySQL/DBeaver) for data auditing, funnel and cohort queries; Excel for LTV/CAC calculations, cohort retention matrices, and MRR breakdowns; and Tableau Public for the interactive dashboard. See Tech Stack and Appendix for further detail.

Funnel Analysis

The user funnel consists of five stages: Signup → Trial → Payment → Renewal → Upgrade.



The largest drop-offs occur at two stages: Signup→Trial (23% of users never start a trial) and Renewal→Upgrade (only 21% of renewing users move to a higher-tier plan).

32% of users who made a first payment (1,150 of 3,602) did so without going through the trial period. This suggests a distinct user path that bypasses the classic Signup→Trial→Payment sequence and may warrant separate analysis (e.g. direct-purchase intent vs. promotional trial-skip offers).

country	t_signed	t_trial	t_first_payment	t_renewal	t_upgrade	Trial→Paid
BR	1000	787	650	396	59	56%
DE	800	595	588	444	98	66%
MX	650	493	414	253	50	53%
PL	450	346	324	180	20	64%
UK	800	603	612	467	93	69%
US	1300	1007	1014	755	215	72%
Total	5000	3831	3602	2495	535	

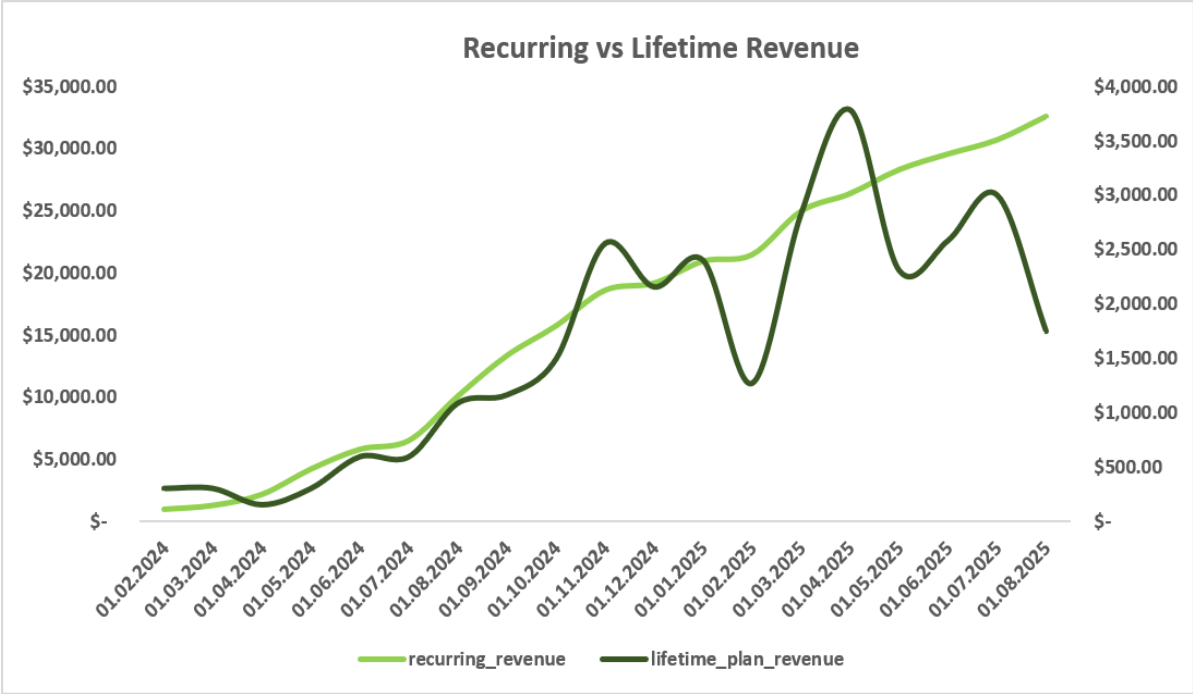
Trial-to-paid conversion varies from 72% in the US to 53% in Mexico. The US, UK and Germany show the highest conversion, while Brazil, Mexico and Poland lag behind, but this comparison should be read with caution, as the app launched later in these markets, giving users in BR/MX/PL less time to reach the later funnel stages (renewal, upgrade). Lower renewal and upgrade counts in these countries may therefore partly reflect a shorter observation window rather than weaker product-market fit.

Methodological note: the Payment figure (3,602) reflects unique users with at least one first payment. The total number of first payment transactions is 3,833; the difference (231) is accounted

for by users with multiple first payment events, most likely representing win-back reactivations after churn.

MRR (Monthly Recurring Revenue)

Revenue is tracked in two components: **recurring revenue** (subscription payments) and **lifetime plan revenue** (one-time lifetime purchases), reported separately to distinguish predictable recurring income from one-off spikes.



Recurring revenue grew consistently from **\$1,020 in February 2024 to \$32,559 in August 2025**, roughly a 32x increase over 19 months, reflecting steady user base growth. Lifetime plan revenue grew more unevenly, from \$300 to \$1,746 over the same period, with several notable spikes tied to specific business events:

- **May 2024:** a jump in revenue coincides with market expansion into the UK and Germany.
- **August 2024:** a further increase follows expansion into Brazil, Mexico, and Poland.
- **November 2024:** both revenue streams spike sharply, driven by Black Friday promotions; the effect is especially pronounced in lifetime plan revenue (\$2,554, more than double the previous month), suggesting lifetime plans were a key part of the Black Friday offer.
- **April 2025:** a smaller spike in lifetime plan revenue aligns with a successful partnership/collaboration in Poland.

Total revenue over the observed period reached **\$312,974** in recurring revenue and **\$30,589** in lifetime plan revenue, confirming that recurring subscriptions are the dominant and more stable revenue driver, while lifetime plans contribute smaller but event-sensitive spikes.

Retention

Retention is measured through monthly cohort analysis, grouping users by signup month and tracking their activity over the following 11 months (M+0 to M+11). Cohorts launched later in the observation period (e.g. 2025-06 to 2025-08) naturally have fewer subsequent months of data available; these cells are intentionally left blank rather than filled with misleading zeros.

Cohort	M+0	M+1	M+2	M+3	M+4	M+5	M+6	M+7	M+8	M+9	M+10	M+11
2024-02	100%	72%	79%	75%	78%	78%	76%	75%	75%	75%	75%	75%
2024-03	100%	60%	67%	66%	66%	66%	66%	64%	63%	63%	63%	63%
2024-04	100%	73%	84%	78%	76%	78%	76%	76%	73%	75%	75%	73%
2024-05	100%	65%	75%	73%	72%	72%	72%	72%	70%	70%	70%	70%
2024-06	100%	72%	79%	75%	74%	75%	75%	75%	75%	74%	74%	73%
2024-07	100%	68%	76%	73%	70%	69%	69%	69%	69%	69%	68%	68%
2024-08	100%	66%	76%	72%	69%	69%	69%	69%	69%	69%	68%	69%
2024-09	100%	61%	72%	68%	68%	68%	67%	67%	66%	66%	66%	66%
2024-10	100%	65%	71%	68%	67%	66%	66%	64%	65%	65%	65%	64%
2024-11	100%	65%	73%	72%	71%	72%	71%	71%	71%	71%	71%	
2024-12	100%	64%	71%	67%	65%	66%	65%	65%	65%	65%		
2025-01	100%	61%	70%	67%	64%	64%	64%	64%	64%			
2025-02	100%	64%	73%	71%	69%	68%	68%	67%				
2025-03	100%	63%	70%	69%	67%	67%	67%					
2025-04	100%	72%	79%	76%	73%	73%						
2025-05	100%	60%	67%	63%	61%							
2025-06	100%	61%	71%	68%								
2025-07	100%	60%	69%									
2025-08	100%	65%										
Average	100%	65%	73%	71%	69%	70%	69%	69%	69%	69%	69%	69%

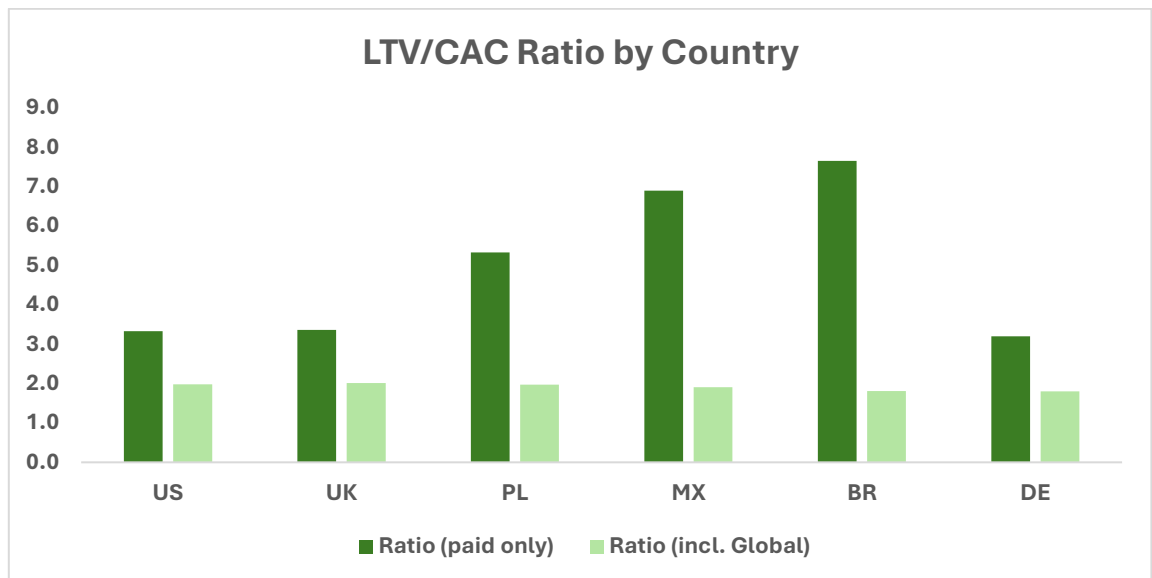
Average retention drops sharply from M+0 (100%) to M+1 (65%), then increases to 73% at M+2, before gradually stabilising around 69-70% from M+4 onward through M+11.

Note on the M+1 to M+2 increase: this is not a data error. Only users who completed a paid action after their initial payment are counted as retained in a given month, and users on annual or lifetime plans are tracked using a snapshot date rather than individual transaction dates, since their ongoing engagement cannot be inferred from transaction history alone. Refunds are not subtracted from retention counts. As a result, some users who show no activity in M+1 reappear as active in M+2, most likely reflecting the natural cadence of billing cycles rather than genuine churn and return.

Payment timing across month boundaries. 16% of first payment transactions (619 of 3,833) occur in a different calendar month than the user's signup month. Of these, 84% are attributable to a trial period that spans across a month boundary, meaning the user started their trial in one month but converted to a paying customer in the next. This is expected behaviour given the trial length and does not indicate a data quality issue.

Overall, retention stabilises at a healthy 69% from month 4 onward, suggesting that users who remain subscribed past the initial drop-off period tend to stay long-term.

Unit Economics (LTV, CAC, ROAS)



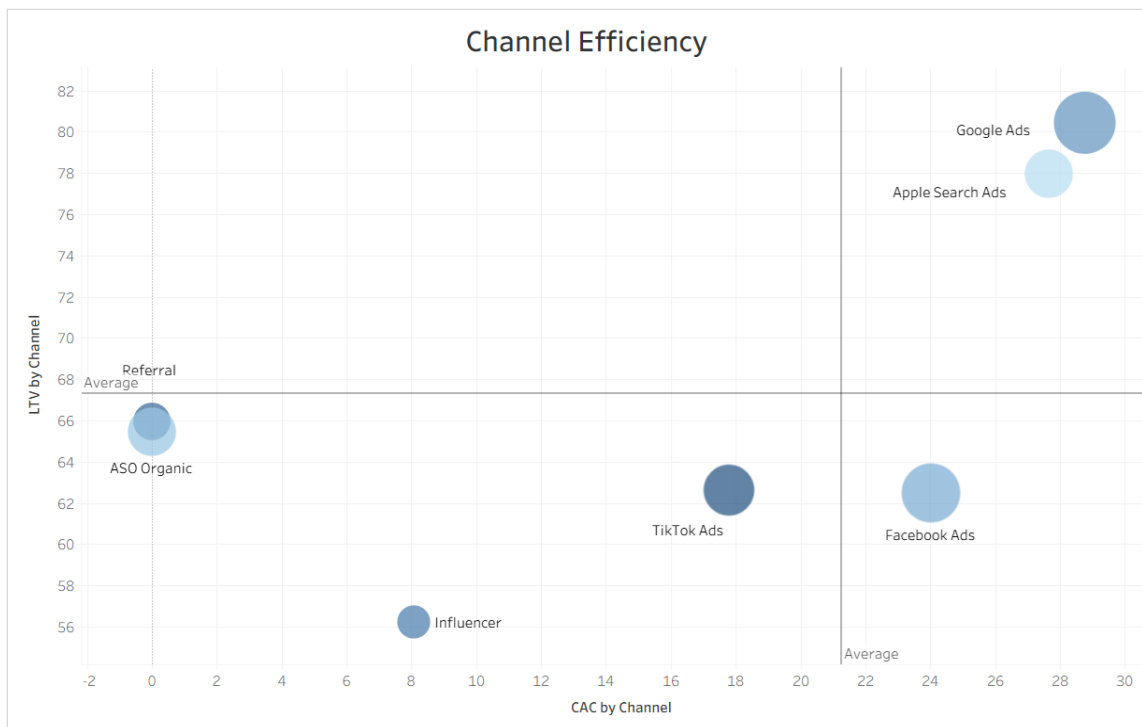
Initial analysis using only paid-channel spend (excluding SMM+Content, which is not attributed to a specific country in the source data) shows a wide spread in LTV/CAC ratio: from 3.2 in Germany to 7.7 in Brazil, despite the US and UK having the highest absolute LTV (\$120.97 and \$124.20 respectively).

However, this comparison excludes \$92,750 in Global (SMM+Content) spend. When this spend is allocated proportionally to each country's share of new users, the picture changes substantially: the ratio converges to a narrow **1.8-2.0 range** across all six countries. This suggests the initial spread was largely an artefact of incomplete cost attribution rather than a genuine difference in channel efficiency between markets.

Note on Global spend allocation: Global (SMM+Content) spend cannot be directly attributed to a specific country in the source data, since it likely covers brand-level campaigns spanning multiple markets. It was allocated proportionally to each country's share of new users as a reasonable approximation, but this remains an estimate rather than a precise attribution and should be treated as a methodological limitation.

Note on LTV calculation: LTV figures are blended, calculated across all users regardless of acquisition channel, including those acquired through free/organic channels (ASO Organic, Referral). LTV/CAC should therefore be read as a general indicator of country-level unit economics rather than a precise return-on-ad-spend metric for paid campaigns specifically.

Channel Efficiency



Influencer marketing has the lowest CAC among paid channels (\$8.39) and the highest LTV/CAC ratio (6.68), despite generating the lowest average LTV (\$56.04). Google Ads and Apple Search Ads, by contrast, have the highest CAC (~\$28) and the highest LTV (\$80.00 and \$77.64 respectively), but the higher spend does not translate proportionally into higher returns.

Referral and ASO Organic are free channels (zero CAC) that still deliver strong LTV (\$65.80 and \$65.15 respectively), making them the most efficient sources of users when available, though their volume (407 and 682 users) is naturally limited by organic reach.

Recommendations

- **Prioritise Influencer campaigns** — lowest CAC among paid channels (\$8.39) and the best LTV/CAC ratio (6.68); scale spend here first.
- **Treat Google Ads and Apple Search Ads as volume, not efficiency, drivers** — they bring the most users and the highest LTV, but at a lower return per dollar. Keep them for reach, not as the primary ROI lever.
- **Keep investing in ASO and Referral** — zero acquisition cost with strong LTV (\$65 range); the cheapest growth available.
- **Fix the Renewal→Upgrade step** — only 21% of renewing users upgrade, the weakest point in the funnel. Test stronger upgrade offers for this already-committed segment.
- **Test a trial-skip offer** — 32% of paying users already convert without a trial. A controlled experiment (e.g. discounted first month instead of a trial) could extend this path to more signups without harming retention.
- **Read country-level LTV/CAC with caution** — Brazil's apparent 7.65 ratio comes from unusually low CAC, not high user value; once Global spend is allocated fairly, all six countries converge to a 1.8–2.0 ratio. Brazil is worth cheap paid growth, not a "best market" label.

Conclusions

This analysis set out to answer two questions: where in the funnel users drop off, and which channels and countries deliver the best return on acquisition spend.

The funnel shows healthy overall conversion (94% trial-to-paid), but two clear weak points: the initial Signup→Trial step and the Renewal→Upgrade step, where only one in five renewing users moves to a higher tier. Retention is stable and strong once past month 4 (69%), indicating that users who stick around tend to stay.

On unit economics, the headline result is that most of the apparent variation in country performance was a data artefact rather than a real signal — once marketing spend is attributed fairly, all markets perform similarly ($LTV/CAC \approx 1.8\text{--}2.0$). Channel-level analysis was more conclusive: Influencer marketing stands out as the most efficient paid channel, while Google Ads and Apple Search Ads remain valuable for volume rather than efficiency.

Overall, the business shows a positive ROAS (1.86) and a growing, increasingly recurring revenue base. The recommendations above focus on the highest-leverage next steps: reallocating some paid budget toward Influencer, addressing the upgrade bottleneck, and testing the trial-skip path already used by a third of paying users.

Tech Stack

SQL (MySQL, via DBeaver) was used for data auditing, funnel and cohort queries. Excel was used for LTV/CAC calculations, cohort retention matrices, and MRR breakdowns. Tableau Public was used to build the interactive dashboard.

Appendix

1. An interactive Tableau Public dashboard:
https://public.tableau.com/views/AppAnalytics_17839450693810/AppAnalytics?:language=en-GB&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link
2. Excel Calculations:
<https://docs.google.com/spreadsheets/d/14s70uGxpltSiXkDrxJnrhJ-IXRI22Zr3/edit?usp=sharing&ouid=102720999807123180118&rtpof=true&sd=true>
3. SQL Queries:
<https://docs.google.com/document/d/1gDrxGhzgNoe6QTYsF2aqZVUU4m9-st0Y/edit?usp=sharing&ouid=102720999807123180118&rtpof=true&sd=true>